

The empirical recurrence for OEIS A225023: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

7 September 2026

Abstract

OEIS A225023 records “Empirical recurrence of order 87”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 489$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 87, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 87 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A225023 is “Number of $(n+1) \times 5$ 0..2 matrices with each 2×2 permanent equal.”. It has offset 1 and begins

2653, 113261, 1930229, 68599847, 1695172731, 52562324371, 1457105103421, 43215008345605, ...

The entry states:

Empirical recurrence of order 87 (see link above)

As of the “Last modified” line on the live entry (Oct 03 2025 23:38:30, revision 9) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 489$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 489$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 978$ exact terms of the model returns order 87 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{87} c_i a(n-i),$$

c_1	55	c_{23}	1013317965322851123129	c_{45}	167010653619577414237741423259
c_2	−486	c_{24}	−6658686818788404332376	c_{46}	59638426363622141148595457449
c_3	−20118	c_{25}	−9153176969473476708225	c_{47}	−46047157905526377599274609297
c_4	385806	c_{26}	92134638829190935443572	c_{48}	115516478352009356037185550650
c_5	1264653	c_{27}	47035813729646297666431	c_{49}	106843849356984364928272195839
c_6	−73042245	c_{28}	−1035790178406437816819082	c_{50}	−54675583703036438745110626082
c_7	214490229	c_{29}	139936209291595644409084	c_{51}	−20749585322395783689031890949
c_8	6868192631	c_{30}	9580672012376794970910421	c_{52}	159622880143826684931430700399
c_9	−43885852516	c_{31}	−6311079256606857623340958	c_{53}	333695517377977125975855069197
c_{10}	−375250247554	c_{32}	−73449571064370312986348646	c_{54}	−351030323327929489831927207416
c_{11}	3811089533795	c_{33}	82252776307711620311772431	c_{55}	−43567073996367238101332495609
c_{12}	11630511211786	c_{34}	468178468843019282010413669	c_{56}	614280060846287012879774607606
c_{13}	−207901779849629	c_{35}	−732511181979334559121817891	c_{57}	443514770451841224648544422761
c_{14}	−100966192604736	c_{36}	−2477938751024055277386102460	c_{58}	−872141490542251811735709711182
c_{15}	7970374768253043	c_{37}	5058430315208466596579459593	c_{59}	−31712084142538201436519387044
c_{16}	−8860155996964724	c_{38}	10808690276496397165869619080	c_{60}	1009584223480037056108247471444
c_{17}	−226171355277329028	c_{39}	−28332936558102362011731383459	c_{61}	937584330475442307164257633674
c_{18}	527513918047878529	c_{40}	−38136365585857450930431871426	c_{62}	−94921665883855372013254510234
c_{19}	4877758704102021534	c_{41}	131452007552302050458959497506	c_{63}	122743104173385578207972041241
c_{20}	−16891897562785265984	c_{42}	103984678525417720473068694187	c_{64}	716335445949752105094784655081
c_{21}	−80700027084585994419	c_{43}	−510682247798970867549939845850	c_{65}	−23637378999793167030736247733
c_{22}	382961769554314121113	c_{44}	−190001218832001662814595713392	c_{66}	−424160539800643851596879132651

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 87, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $87 + 4$ terms removed returns order 87 as well, so $\deg q_{\min} = 87 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once

more on the sequence with its first $87 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 87 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A225023.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.